Lecture 6. Nonlinear Equations

Bisection, fixed-point iteration, contraction mapping, Newton's method

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We seek x with $f(x) = 0$ for nonlinear f; there is no direct method, so we iterate. A critical point of g is a root of $f = g'$ (or ∇g), so root finding covers maximum likelihood, nonlinear least squares, Bayes and MAP estimation. Examples: a root of $f(x) = x^4 - \sin x$; the maximizer of $g(x) = \log x/(1+x)$, a root of $g'(x) = \{1 + x^{-1} - \log x\}/(1+x)^2$.

6.1 Bisection

If f is continuous on $[l, r]$ with $f(l)f(r) \le 0$, there is a root in $[l, r]$. At $m = (l+r)/2$: stop if $f(m) = 0$; else replace r by m if $f(l)f(m) < 0$, and l by m otherwise. After n steps the bracket has length $2^{-n}(r^{(0)} - l^{(0)})$. Bisection gains one bit per step but always works. Typical use: quantiles, $F(x) - u = 0$ for a continuous cdf F; if $F = \int_{-\infty}^{x} f$ needs numerical integration (Lecture 8), this gives the inverse cdf method of Lecture 9 when F^{-1} has no closed form.

6.2 Fixed-point iteration

x is a **fixed point** of F if $F(x) = x$. Choose F with $f(x) = 0 \iff F(x) = x$ and iterate $x^{(k+1)} = F(x^{(k)})$ (Lecture 5 is the case $F(x) = Rx + c$).

Theorem 6.1 (Contraction mapping)

Let $I \subset \mathbb{R}$ be a closed interval and $F : I \to I$ with $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$, where $0 \le L < 1$. Then F has a unique fixed point $x^\star \in I$, the iteration converges to it from every $x^{(0)} \in I$, and

$$|x^{(n)} - x^\star| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|.$$

Proof. $|x^{(k+1)} - x^{(k)}| = |F(x^{(k)}) - F(x^{(k-1)})| \le L|x^{(k)} - x^{(k-1)}| \le \cdots \le L^k|x^{(1)} - x^{(0)}|$. For $m > n$, by the triangle inequality,

$$|x^{(m)} - x^{(n)}| \le \sum_{k=n}^{m-1} L^k|x^{(1)} - x^{(0)}| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|. \qquad (*)$$

So the sequence is Cauchy, with limit $x^\star \in I$ (I closed), and continuity gives $x^\star = \lim F(x^{(k)}) = F(x^\star)$. Another fixed point $y^\star$ would satisfy $|x^\star - y^\star| \le L|x^\star - y^\star|$, so $x^\star = y^\star$. Let $m \to \infty$ in $(*)$. ■

The geometric rate L^n is **linear convergence**.

Proposition 6.2

If F is differentiable on $I = [a, b]$, then $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$ if and only if $|F'(x)| \le L$ for all $x \in I$. Hence the best Lipschitz constant is $\sup_I |F'|$.

Proof. ($\Rightarrow$) Difference quotients at x are bounded by L, hence so is their limit $F'(x)$.

($\Leftarrow$) Mean value theorem: $|F(x) - F(y)| = |F'(c)||x - y| \le L|x - y|$. ■

Example 6.1 (Relaxed fixed-point iteration)

To solve $f(x) = 0$, iterate $F(x) = x + \alpha f(x)$, $\alpha \ne 0$. By Proposition 6.2 and Theorem 6.1 it converges if F maps an interval I around the root into itself and $\sup_I |1 + \alpha f'| < 1$. This needs f' of constant sign; if $0 < m \le f' \le M$ on I, every $\alpha \in (-2/M, 0)$ works and

$\alpha = -2/(m+M)$ gives the best $L = (M-m)/(M+m)$. Newton's method is the adaptive choice $\alpha = -1/f'(x^{(t)})$, making $F'(x^\star) = 0$. For $f = g'$ and $\alpha = -\gamma$ this is gradient descent $x^{(t+1)} = x^{(t)} - \gamma g'(x^{(t)})$ (Lecture 7), and the condition is the step-size bound $\gamma < 2/\sup g''$.

Example 6.2

For $f(x) = x^4 - \sin x$ on $(0, 1]$, both $F_1(x) = (\sin x)^{1/4}$ and $F_2(x) = \sin x / x^3$ fix the root. F_1 maps $I = [0.2, 1]$ into itself and $F_1'(x) = \cos x / \{4(\sin x)^{3/4}\}$ is decreasing with $0 < F_1'(1) < F_1'(0.2) < 1$: a contraction, so the iteration converges. F_2 maps $[0.9, 1]$ onto $[0.841, 1.075]$ with $F_2'(1) < -1$: it expands and diverges. The choice of F matters.

6.3 Newton's method

Motivation. Newton's method replaces f by its tangent, doubling the correct digits near a simple root, at the price of derivatives and possible divergence.

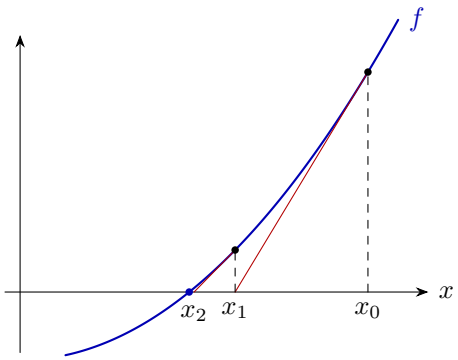


Figure. *Newton's method:* $x_{k+1} = x_k - f(x_k)/f'(x_k)$ *is the zero of the tangent at* x_k.

Linearizing, $0 = f(x^\star) \approx f(x^{(k)}) + f'(x^{(k)})(x^\star - x^{(k)})$, suggests

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

i.e. fixed-point iteration with $F(x) = x - f(x)/f'(x)$, for which

$$F'(x) = \frac{f(x)f''(x)}{f'(x)^2}.$$

Theorem 6.3 (Local quadratic convergence)

Let $f(x^\star) = 0$, $f'(x^\star) \neq 0$, and f be three times continuously differentiable near $x^\star$. Then for $x^{(0)}$ close enough to $x^\star$, Newton's iterates converge to $x^\star$ and

$$\lim_{k\to\infty} \frac{|x^{(k+1)} - x^\star|}{|x^{(k)} - x^\star|^2} = \frac{1}{2}|F''(x^\star)| = \frac{|f''(x^\star)|}{2|f'(x^\star)|}.$$

Proof. $F'(x^\star) = 0$, so $|F'| \le \frac{1}{2}$ on a closed neighbourhood I of $x^\star$, which F maps into itself; Theorem 6.1 gives convergence. By Taylor,

$$x^{(k+1)} - x^\star = F(x^{(k)}) - F(x^\star) = F'(x^\star)(x^{(k)} - x^\star) + \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2 = \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2,$$

$\tilde{x}$ between $x^{(k)}$ and $x^\star$. Divide and let $k \to \infty$; differentiating F' gives $F''(x^\star) = f''(x^\star)/f'(x^\star)$. ■

Quadratic convergence roughly doubles the correct digits per step, but only *locally*: from a poor start Newton's method can wander, reach an unintended root, or diverge (e.g. roots of $\cos x$). In practice it is safeguarded by bisection or line search.

6.4 Systems of equations

For $f : \mathbb{R}^n \to \mathbb{R}^n$ with Jacobian $Jf(x) = (\partial f_i/\partial x_j)$, Newton's method is

$$x^{(k+1)} = x^{(k)} - [Jf(x^{(k)})]^{-1} f(x^{(k)}),$$

solving $Jf(x^{(k)})\Delta x = -f(x^{(k)})$ (e.g. by LU), never inverting. Cheaper variants replace the Jacobian:

- modified Newton: the fixed $J = Jf(x^{(0)})$ (factor once);
- Jacobi: $\operatorname{diag} Jf(x^{(k)})$;
- Gauss–Seidel: the lower triangle of $Jf(x^{(k)})$;
- SOR: $\frac{1}{\omega}\operatorname{diag} Jf(x^{(k)})$ plus the strict lower triangle.

They converge more slowly (typically linearly) with much cheaper steps.

6.5 Selected exercises

Motivation. Global convergence for convex increasing f, the multiple-root case, and an application with exact error squaring.

Exercise 6.1 (Global convergence for convex increasing functions)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable and convex with $f' > 0$, and let $f(x^\star) = 0$.
(a) Show that from every $x^{(0)}$ the Newton iterates satisfy $x^{(k)} \geq x^\star$ for $k \geq 1$, decrease monotonically from $k = 1$ on, and converge to $x^\star$.
(b) Apply this to compute $\log a$ $(a > 0)$ without logarithms.

Solution. [Supplement]

(a)

Step 1. (Overshoot.) Convexity gives $f(y) \geq f(x) + f'(x)(y - x)$; with $x = x^{(k)}$, $y = x^{(k+1)}$ the right side is 0, so $f(x^{(k+1)}) \geq 0 = f(x^\star)$, and $x^{(k+1)} \geq x^\star$ for $k \geq 0$ as f is strictly increasing.

Step 2. (Monotonicity.) For $k \geq 1$, $f(x^{(k)}) \geq 0$ and $f'(x^{(k)}) > 0$, so $x^{(k+1)} \leq x^{(k)}$.

Step 3. (Limit.) $(x^{(k)})_{k\geq 1}$ is nonincreasing and bounded below by $x^\star$, so $x^{(k)} \to \ell \geq x^\star$. By continuity of f, f' and $f'(\ell) > 0$, $\ell = \ell - f(\ell)/f'(\ell)$, so $f(\ell) = 0$ and $\ell = x^\star$ (unique root).

(b) $f(x) = e^x - a$ is convex, $f' = e^x > 0$, root $\log a$:

$$x^{(k+1)} = x^{(k)} - \frac{e^{x^{(k)}} - a}{e^{x^{(k)}}} = x^{(k)} - 1 + a\, e^{-x^{(k)}},$$

converging to $\log a$ from every start by (a), quadratically by Theorem 6.3 ($f'(\log a) = a \neq 0$). ■

Exercise 6.2 (Newton's method at a multiple root)

Let $f(x) = (x - x^\star)^m g(x)$ with an integer $m \geq 2$, g continuously differentiable near $x^\star$ and $g(x^\star) \neq 0$.
(a) Show that Newton's method converges to $x^\star$ from every point close enough to $x^\star$, but only linearly: $\lim_k (x^{(k+1)} - x^\star)/(x^{(k)} - x^\star) = (m-1)/m$.
(b) Show that the modified iteration $x^{(k+1)} = x^{(k)} - m f(x^{(k)})/f'(x^{(k)})$ satisfies $|x^{(k+1)} - x^\star| \leq C|x^{(k)} - x^\star|^2$ near $x^\star$.

Solution. [Supplement] Write $e = x - x^\star$. Then $f'(x) = e^{m-1}\{mg(x) + eg'(x)\}$ with bracket $\to mg(x^\star) \neq 0$, so nonzero on some $|e| < \delta_0$, where $f'(x) \neq 0$ for $x \neq x^\star$ and

$$\frac{f(x)}{f'(x)} = \frac{e\, g(x)}{mg(x) + eg'(x)}.$$

(An iterate equal to $x^\star$ stops at the root.)

(a) The Newton map satisfies

$$F(x) - x^\star = e\Big(1 - \frac{g(x)}{mg(x) + eg'(x)}\Big) = e\,\varphi(x), \qquad \varphi(x) = \frac{(m-1)g(x) + eg'(x)}{mg(x) + eg'(x)}.$$

φ is continuous with $\varphi(x^\star) = (m-1)/m < 1$, so $|\varphi| \leq q < 1$ on some $|e| < \delta \leq \delta_0$. There $|e_{k+1}| \leq q|e_k|$, the iterates stay and $e_k \to 0$, and $e_{k+1}/e_k = \varphi(x^{(k)}) \to (m-1)/m$.

(b) For $G(x) = x - mf(x)/f'(x)$,

$$G(x) - x^\star = e\Big(1 - \frac{mg(x)}{mg(x) + eg'(x)}\Big) = e^2\,\psi(x), \qquad \psi(x) = \frac{g'(x)}{mg(x) + eg'(x)},$$

with ψ continuous, hence bounded by some C on a closed interval around $x^\star$. So $|e_{k+1}| \leq Ce_k^2$: quadratic convergence for $C|e_0| < 1$ (e_0 in that interval). This needs m; otherwise apply Newton to f/f', which has a simple root at $x^\star$. ■

Exercise 6.3 (Division by multiplication)

Let $a > 0$ and apply Newton's method to $f(x) = 1/x - a$.
(a) Show that the iteration is $x^{(k+1)} = x^{(k)}(2 - ax^{(k)})$, which uses no division.
(b) With $r_k = 1 - ax^{(k)}$, show that $r_{k+1} = r_k^2$.
(c) Conclude that $x^{(k)} \to 1/a$ if and only if $0 < x^{(0)} < 2/a$, that the convergence is then quadratic, and describe what happens otherwise.

Solution. [Supplement]

(a) $f'(x) = -1/x^2$, so $x - f(x)/f'(x) = x + x^2(1/x - a) = 2x - ax^2$.

(b) $1 - ax(2 - ax) = 1 - 2ax + a^2x^2 = (1 - ax)^2$.

(c) By (b), $r_k = r_0^{2^k}$ and $x^{(k)} = (1 - r_k)/a$. If $|r_0| < 1$, i.e. $0 < x^{(0)} < 2/a$, then $x^{(k)} \to 1/a$ with

$$\Big|x^{(k+1)} - \frac{1}{a}\Big| = \frac{r_k^2}{a} = a\Big|x^{(k)} - \frac{1}{a}\Big|^2,$$

so the correct bits double exactly. If $|r_0| = 1$ ($x^{(0)} \in \{0, 2/a\}$), $x^{(k)} = 0$ for $k \geq 1$; if $|r_0| > 1$, $x^{(k)} \to -\infty$. Division routines use this iteration after a table lookup gives a start with small $|r_0|$. ■